Implementation of the Dehbi (2008) CRW–Langevin Particle Dispersion Model as an ANSYS Fluent UDF

Technical report: 90° pipe-bend deposition case

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Abstract

The continuous random walk (CRW) model of Dehbi (2008) was implemented as a compiled ANSYS Fluent user-defined function (UDF) and applied to turbulent aerosol deposition in a 90° pipe bend at $Re \approx 4.1 \times 10^3$. The UDF integrates a normalised Langevin equation for the fluid velocity seen by each particle, with anisotropic near-wall statistics and a wall-normal drift correction, and couples it to Fluent's discrete phase model through a body force. On a Reynolds-stress-model flow field, the computed deposition fraction rises from 16.7 % at a flow Stokes number of 0.09 to 98.0 % at 2.3. A line-by-line review and a statistical test of the random-number generator identify three aspects of the implementation that define the scope of the present results.

1. Introduction

Discrete random walk (DRW) models sample velocity fluctuations from locally isotropic turbulence. In inhomogeneous near-wall turbulence they produce a spurious drift of low-inertia particles towards the wall, violating the well-mixed condition and over-predicting small-particle deposition. Dehbi (2008) replaced the DRW with a normalised Langevin equation that uses anisotropic near-wall rms profiles and a drift term proportional to the wall-normal gradient of the rms velocity. This work implements that model in ANSYS Fluent and computes the deposition fraction in a 90° bend for particle diameters of 10 to 50 µm.

2. Case set-up

Table 1. Geometry, flow conditions and solver settings.

Item	Value
Pipe diameter D; bend curvature ratio R/r	0.02 m; 4
Straight inlet and outlet sections	$5D$ each
Fluid	air, ρ = 1.225 kg m^{-3}, μ = 1.81 × 10^{-5} Pa s
Inlet; outlet	uniform velocity 3.0 m s^{-1}; pressure outlet
Reynolds number $\rho UD/\mu$; friction Reynolds number $u_\tau r/\nu$	4.06 × 10^3; ≈ 140 (Blasius estimate)
Mesh	structured hexahedral, ≈ 5.3 × 10^6 cells, wall $y^+ \approx 1$
Solver	ANSYS Fluent 2024 R2, pressure based, steady, SIMPLE
Turbulence model	Reynolds stress model, linear pressure–strain, enhanced wall treatment
Discretisation	least-squares gradients, second-order pressure, second-order upwind
Discrete phase	inert spheres, one-way coupling, steady tracking, 10,000 particles per diameter, built-in dispersion disabled
Deposition	particle stopped at first wall contact (reported by Fluent as "aborted")



Figure 1. Computational domain.

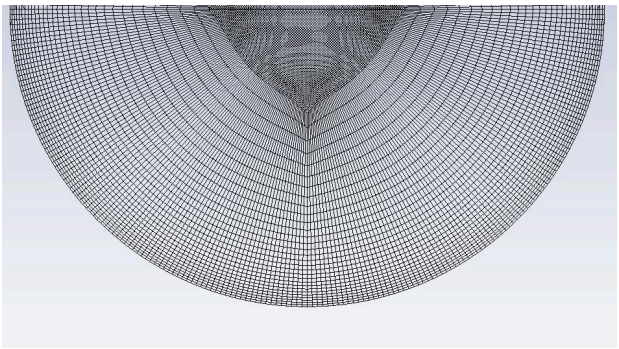
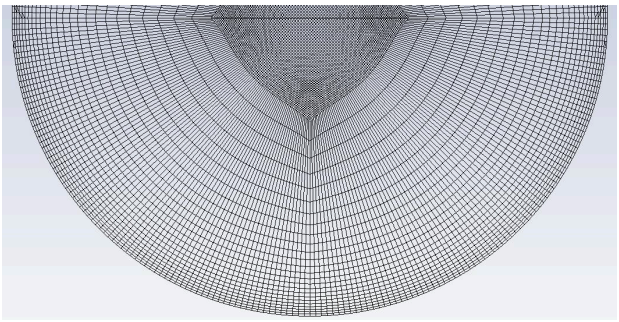
Refined mesh, 5.3 × 10^6 cells	**Baseline mesh, 2.5 × 10^6 cells**
	

Figure 2. Bend cross-section of the two meshes. Small-particle deposition was sensitive to the near-wall resolution; the refined mesh was used for all results.

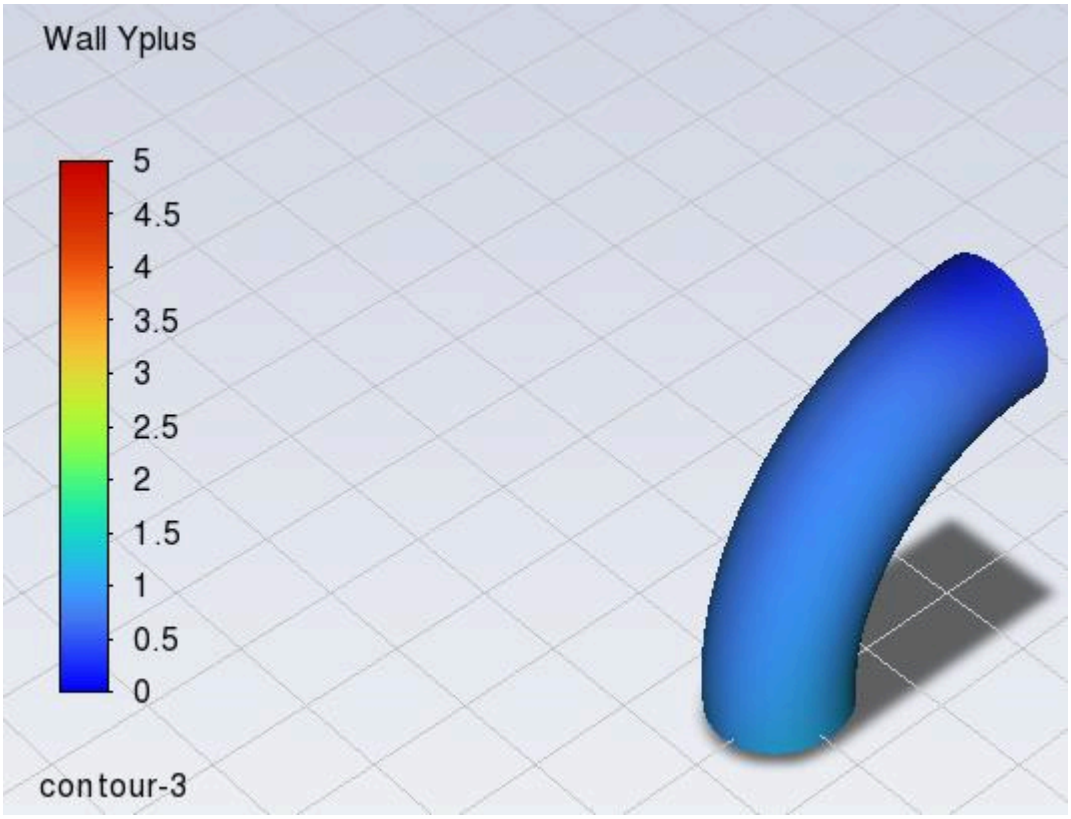


Figure 3. Wall y^+ on the bend surface.

The scaled residuals of the continuous phase decreased to between 10^{-3} and 2×10^{-6} (`figures/residuals.png`). The Fluent dialogs for the turbulence model, solution methods, discrete-phase numerics, UDF hooks, injection and compilation are archived in `figures/setup/`.

3. Model implementation

Line numbers refer to `udf/crw_dehbi2008_rev10.c`.

Coupling. Fluent integrates the particle momentum equation with the mean slip velocity. The UDF adds the drag of the fluid velocity fluctuation u'_i as a body force per unit mass (lines 644–738):

$$\frac{F_i}{m_p} = f_D \, u'_i, \qquad f_D = \frac{1 + 0.15\, Re_p^{0.687}}{\tau_p}, \qquad \tau_p = \frac{\rho_p d_p^2}{18\,\mu}.$$

Langevin equation. The normalised fluctuation $\eta_i = u'_i/\sigma_i$ obeys $d\eta_i = -\eta_i\, dt/\tau_L + A_i\, dt + \sqrt{2/\tau_L}\, dW_i$ and is advanced with a semi-implicit Euler–Maruyama scheme (lines 397–399, 427–429):

$$\eta_i^{n+1} = \frac{\eta_i^n + \sqrt{2\,\Delta t/\tau_L}\;\xi_i + A_i\,\Delta t}{1 + \Delta t/\tau_L}.$$

Near-wall branch ($y^+ < 100$). The equation is solved in a body-fitted frame (wall normal, mean-flow direction, binormal; lines 225–275). The rms profiles are the DNS fits given by Dehbi (2008) (lines 131–148):

$$\frac{\sigma_1}{u_\tau} = \frac{0.40\, y^+}{1 + 0.0239\,(y^+)^{1.496}}, \quad \frac{\sigma_2}{u_\tau} = \frac{0.0116\,(y^+)^2}{1 + 0.203\, y^+ + 0.00140\,(y^+)^{2.421}}, \quad \frac{\sigma_3}{u_\tau} = \frac{0.19\, y^+}{1 + 0.0361\,(y^+)^{1.322}}.$$

The Lagrangian time scale follows Kallio and Reeks (1989): $\tau_L^+ = 10$ for $y^+ \le 5$ and $\tau_L^+ = 7.122 + 0.5731\, y^+ - 0.00129\,(y^+)^2$ otherwise (lines 196–214). Only the wall-normal component carries a drift, reduced by the local particle Stokes number $St = \tau_p/\tau_L$; its derivative is evaluated analytically (lines 163–188, 421–422):

$$A_2 = \frac{1}{1+St}\frac{\partial \sigma_2}{\partial y}, \qquad A_1 = A_3 = 0.$$

Bulk branch ($y^+ \geq 100$). Isotropic turbulence with $\sigma = (2k/3)^{1/2}$ and $\tau_L = (2/C_0)\,k/\varepsilon$, $C_0 = 14$ (lines 444–474). The ∇k drift term is omitted, because accessing the cell gradient of k under the RSM caused a solver failure.

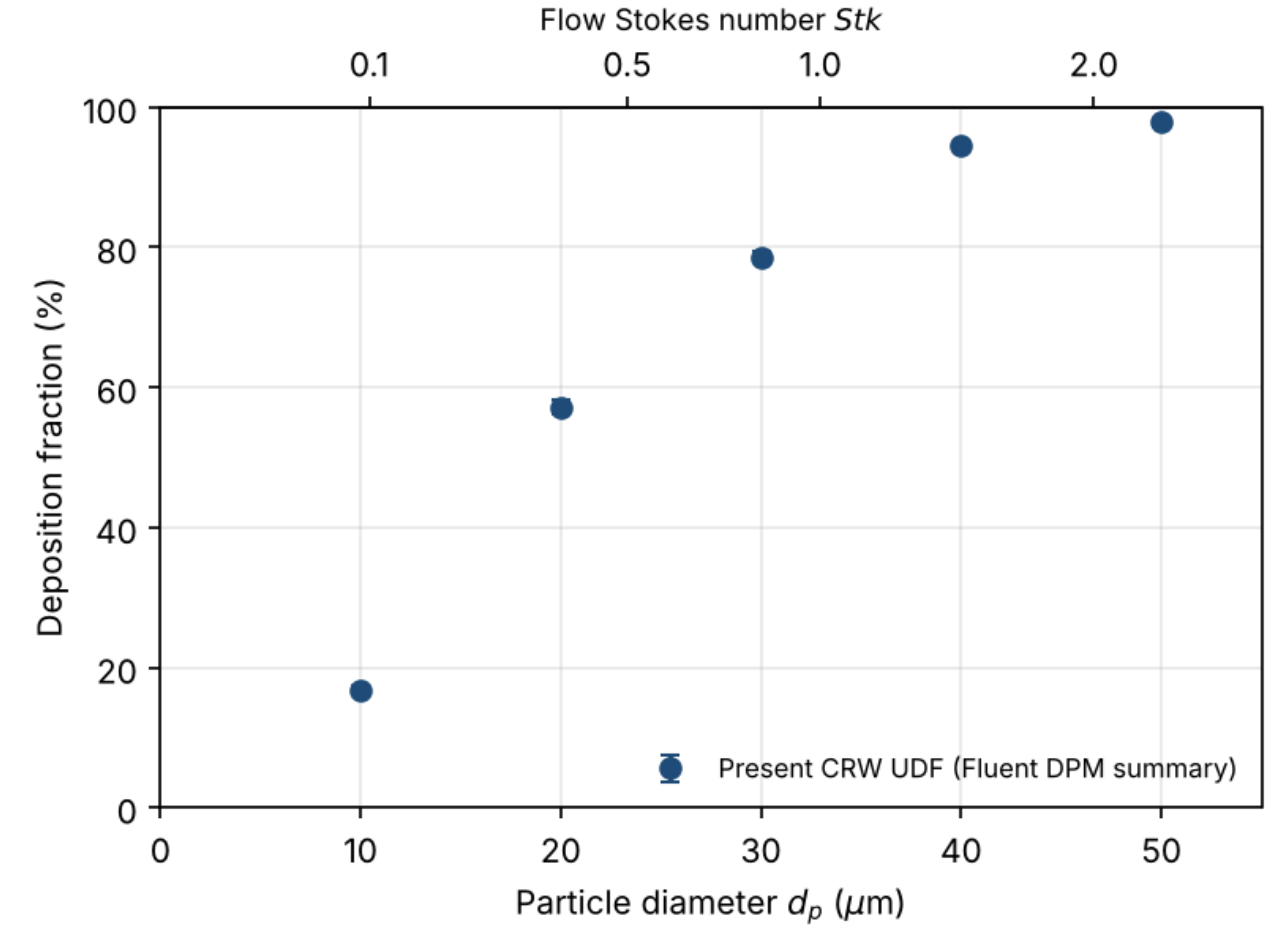
Wall data and time step. The on-demand function `init_wall_udm` (lines 498–625) stores u_τ, y^+ and the wall normal in user-defined memory. The particle time step is $\min(10^{-6}\ \text{s},\ 0.1\,\tau_p)$ (lines 765–797).

4. Results

The flow Stokes number is $Stk = \rho_p d_p^2 U/(9\mu D)$, evaluated with ρ_p = 1000 kg m^{-3}. The standard error is $\sqrt{DF(1-DF)/N}$.

Table 2. Deposition fraction (DF) from the Fluent DPM summaries (`data/deposition_fraction.csv`).

d_p (µm)	*Stk*	Tracked	Escaped	Deposited	DF (%)	Standard error (%)
10	0.092	10,000	8,329	1,671	16.7	0.37
20	0.368	10,000	4,279	5,721	57.2	0.49
30	0.829	10,000	2,142	7,858	78.6	0.41
40	1.473	10,000	546	9,454	94.5	0.23
50	2.302	10,000	199	9,801	98.0	0.14



10,000 particles per diameter. 95% sampling intervals are smaller than the markers.

Figure 4. Deposition fraction against particle diameter and flow Stokes number, generated by `tools/plot_deposition.py`.

The deposition fraction increases monotonically with Stokes number, from inertia-limited deposition at $Stk < 0.1$ to near-complete impaction at $Stk > 1.5$, which is the expected qualitative behaviour for bend flows. Deposition is counted on all wall zones.

5. Code verification

Random-number generator. The function `gauss_rand` (lines 94–121) generates Box–Muller deviates from a hash of the particle address and the particle time, without a stored state. The program `tools/rng_check.c` runs an exact copy of it.

Table 3. Statistics of the UDF generator at $\Delta t = 10^{-6}$ s.

Statistic	Ideal	Result
Mean, variance (2×10^5 samples)	0, 1	+0.001, 1.003
Autocorrelation, lags 1 to 5	0	+0.22, +0.06, −0.11, −0.29, −0.23
Variance of 1000 accumulated deviates divided by 1000, 2000 particle addresses	1	median 0.48; 5–95 % range 0.20 to 2.7

The one-point statistics are correct, while successive deviates are correlated, so the effective turbulent diffusivity varies between particles.

Implementation scope. The review also establishes that:

1. the near-wall statistics are applied in the wall-adjacent cell layer and evaluated at the cell centre; other cells use the bulk branch;
2. the Langevin state is stored in the body-fitted frame near the wall and in Cartesian components in the bulk, without rotation between the two.

6. Limitations and further work

The present case differs from the turbulent bend experiment of Pui et al. (1987), reported as $D = 8.51$ mm, curvature ratio 5.6 and $Re = 10{,}000$ (Sakib et al., 2023); a comparison with that benchmark is meaningful only at matched Stokes number. The $5D$ inlet section does not produce fully developed flow at the bend. The archived injection dialog for the smallest size class specifies a diameter of 5 µm and the material anthracite, whereas Table 2 assumes 10 µm and 1000 kg m^{-3}, and the gravity setting is not recorded in the archived dialogs. Further work comprises a per-particle generator state, near-wall statistics evaluated at the particle position across the whole $y^+ < 100$ region, a consistent frame transformation, a well-mixed tracer test, and a comparison with the Pui et al. data at matched Stokes number.

7. Conclusions

A CRW–Langevin dispersion model following Dehbi (2008) was implemented as a compiled ANSYS Fluent UDF with anisotropic near-wall statistics, a body-fitted frame, a Stokes-number-weighted drift correction and an adaptive particle time step. Applied to a 90° bend, it gives deposition fractions from 16.7 % to 98.0 % over $0.09 \le Stk \le 2.3$. The accompanying verification tools quantify the statistical properties of the random forcing and define the improvements required for a quantitative benchmark comparison.

Acknowledgement of software tools

The Cursor editor was used to debug C compilation errors during UDF development.

References

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